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BHOPAL

Queuing Models & Second Order Cone Optimization

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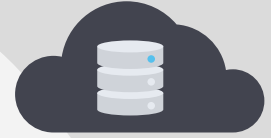




Part 1

Basic Components of a Queuing System

Overview of a Queuing System



What is a Queue?

A queuing system models the flow of customers arriving at a service facility, waiting in line, and being served. It consists of an input process (arrivals), a waiting line (queue), and a service mechanism. Queuing theory helps analyze congestion, optimize resource allocation, and improve system efficiency across industries such as telecommunications, healthcare, and manufacturing.

Kendall's Notation

Queuing systems are classified using Kendall's notation: $A/S/c/K/N/D$, where A = arrival distribution, S = service time distribution, c = number of servers, K = system capacity, N = population size, and D = queue discipline. Common models include $M/M/1$ (single server, Poisson arrivals, exponential service) and $M/M/c$ (multi-server variants).

Arrival Process



Poisson Arrivals

The most common arrival model assumes customers arrive according to a Poisson process with mean rate λ . Arrivals are random, memoryless, and independent of each other.

Batch Arrivals

In some systems, customers arrive in groups (bulk arrivals). Examples include buses delivering passengers or packets arriving in bursts in networks.

Interarrival Time

The time between consecutive arrivals. For Poisson arrivals, interarrival times follow an exponential distribution with mean $1/\lambda$, enabling mathematical tractability.

Arrival Rate λ

The average number of customers arriving per unit time. It is a key parameter in Little's Law: $L = \lambda W$, linking system population, arrival rate, and average wait time.





Service Mechanism



Service Time

The duration a server spends on each customer. In the M/M/1 model, service times follow an exponential distribution. General distributions (G) are used in more complex models like M/G/1.

Number of Servers

Systems may have single ($c=1$) or multiple ($c>1$) parallel servers. Multi-server models like M/M/c reduce wait times by sharing the load across servers simultaneously.

Service Rate μ

The average number of customers a server can handle per unit time (μ). Traffic intensity $\rho = \lambda/(c\mu)$ must be less than 1 for a stable system — otherwise queues grow unboundedly.



Queue Discipline & System Structure

FCFS

First-Come, First-Served: customers are served in order of arrival. The most common and fairest queue discipline used in real-world systems.

LCFS

Last-Come, First-Served: the most recently arrived customer is served first. Common in stack-based systems like memory management and inventory.

Priority Queues

Higher-priority customers are served first. Used in emergency rooms, OS scheduling, and network traffic management.

Round Robin

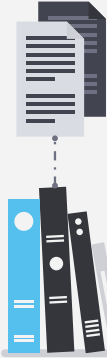
Each customer gets a fixed time slice in rotation. Common in CPU scheduling where equal fairness across all users is required.

System Capacity

Max customers allowed in the system. When capacity K is reached, new arrivals are turned away (finite buffer models).

Calling Population

Infinite (open) or finite (closed) source of customers. E.g., internet users vs. machines served by a single repair team.



Departure Process

Definition

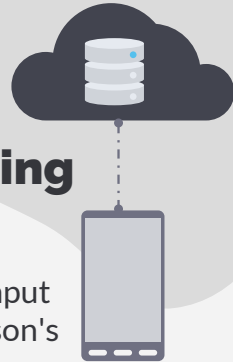
The departure process describes how customers/jobs exit the queuing system after service completion. The departure rate equals the arrival rate λ in steady state (throughput = λ when $\rho < 1$). Key Properties In an M/M/1 queue, departures follow a Poisson process with rate λ (Burke's Theorem) Departure rate = $\min(\lambda, \mu)$ — server cannot output faster than it processes In unstable systems ($\rho \geq 1$), effective throughput = μ (server always busy)

Burke's Theorem

For M/M/1, M/M/c, and M/M/ ∞ queues in steady state: The departure process is Poisson with rate λ The number in system at time t is independent of departure times before t This enables analysis of tandem/network queues

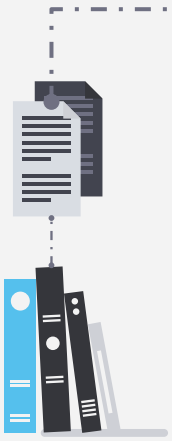
Departure in Queuing Networks

Output of one queue becomes input (arrival) to the next queue Jackson's Theorem: each node in an open network behaves independently as M/M/c Used in ML pipeline modeling: preprocessing $\rightarrow$ inference $\rightarrow$ post-processing



Key Formula Effective throughput:

$$\Lambda = \lambda(1 - P_{\text{block}}) \text{ (for finite capacity systems)}$$



$$L = \lambda W$$

Little's Law: Avg. customers in system = arrival rate $\times$ avg. time in system

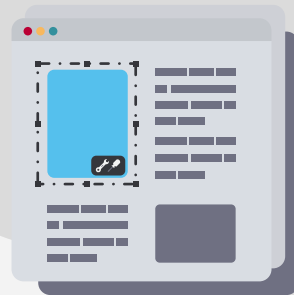


$$\rho = \lambda / (c\mu)$$

Traffic Intensity: must remain below 1 for queue stability

$$Wq = Lq / \lambda$$

Avg. waiting time in queue: customers waiting before service begins



Part 2

Second Order Cone Optimization

What is Second Order Cone Programming?



SOCp Definition

Second Order Cone Programming (SOCp) is a class of convex optimization where the feasible region is defined by second-order cone (SOC) constraints of the form $\|Ax + b\| \leq c^T x + d$. It generalizes linear programming (LP) and encompasses quadratic programming (QP), making it extremely powerful for modeling queuing systems with stochastic constraints.

SOCp in Queuing Theory

SOCp enables optimization of queuing systems under uncertainty. Key applications include minimizing average waiting time subject to server capacity constraints, robust scheduling with probabilistic service-level guarantees, and optimal resource allocation across multi-server networks where variance bounds are expressed as SOC constraints.

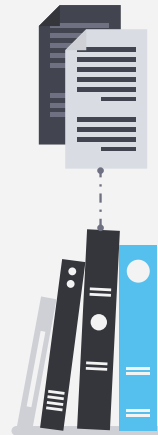
SOCP Standard Form

Primal Formulation

$\min f^T x$ subject to $\|A_i x + b_i\|_2 \leq c_i^T x + d_i$
 $Fx = g$ (linear equality constraints)
 $x \in \mathbb{R}^n$ (decision variable vector)

Second Order Cone

$Q_+^{n+1} = \{(x, t) : \|x\|_2 \leq t\}$ (Lorentz cone)
Generalizes LP when cone constraints are linear
Includes QCQP as special case with quadratic terms
Solved efficiently by interior-point solvers (CVXPY, Mosek)





SOCP Applications in Queuing Systems

Robust Queue Optimization

SOCP formulates worst-case waiting time minimization under uncertain arrival rates. Chance constraints like $P(W_q \leq T) \geq 1 - \epsilon$ are reformulated as tractable SOC constraints using moment information.

Network Flow Control

In queueing networks (e.g. Jackson networks), SOCP controls routing probabilities and buffer sizes to minimize end-to-end latency while satisfying throughput and stability constraints at every node.

Server Allocation

Optimally assign servers across parallel queues to minimize total delay subject to staffing budget constraints. SOCP solves this efficiently even when service variance is included in the model.

Stochastic Scheduling

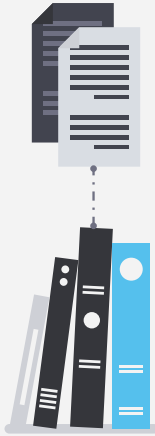
Job scheduling under random processing times is cast as an SOCP by bounding the standard deviation of completion times. This yields robust schedules with provable delay guarantees across all realizations.



Conclusion

Queuing Models provide a mathematical framework to analyze systems where arrivals compete for limited service resources. The four core components — Arrival Process (λ), Waiting Queue, Service Mechanism (μ , c), and Departure Process — along with performance metrics like utilization ρ , queue lengths, waiting times, and Little's Law ($L = \lambda W$), enable systematic evaluation and optimization of real-world systems. Second Order Cone Programming (SOCP) complements this by offering a powerful convex optimization framework that generalizes LP and QP through Lorentz cone constraints, solvable in polynomial time via interior-point methods, with broad applications in signal processing, machine learning, and finance.

Together, these two topics form a strong analytical foundation — queuing theory addressing stochastic resource contention and SOCP providing structured deterministic optimization — both essential tools in the design and performance tuning of modern computational systems.





THANK YOU

